

10/5/24

- 1) Machine Learning and Deep Learning
Deep Learning branch Artificial intelligence DL is ~~learn~~ learn
 Neural Network blueprint A human brain structure is learn
 Unstructured data
Machine Learning is learn from data is use structured
 data only is mainly used supervised and unsupervised
 learning and Reinforcement learning for improve performance

	ML	DL
Architecture	single algorithm used	multiply layer neuron networks used
data	less data used	large amount data used
Feature	manually handle process	Automation working based instructions
application	Structure data	speech, video, text

Example:

ML :- spam detection, fraud detection

DL :- face recognition,

2) Explain the concept of Representation Learning. Discuss how the width and depth of a Neural Network.

Representation Learning allow a Neural Network to learn Feature set automatically from raw data.

Width of a Neural Network

- Refers to the number of neurons in a layers
- improve Feature Learning
- Increase memory usage

Depth of a Neural Network

- Refers to the Number of hidden layers.
- Learn complex patterns
- ~~More~~ $\uparrow$
- Increases computational cost.

3. Compare ReLU, Leaky, ReLU (LReLU) and ELU (Exponential Linear Unit).

Functions :- ~~Formula~~

ReLU

Formula :- $f(x) = \max(0, x)$

Advantages :- Fast computation

disadvantage :- Dead neuron problem

Leaky ReLU

Formula :- $f(x) = x, 0.01x$

Advage :- Reduce dead neurons

disadvantage :- Higher Complexity

Feature	RBM	Autoencoder
Objective	Feature Extraction	Data reconstruction

ELU :-

Formula :- $f(x) = \alpha(e^x - 1)$

18

Advantage :- faster convergence

Disadvantage :- computationally expensive

Applications

- ReLU - Image Classification
- ~~ReLU~~
- Leaky ReLU - Object detection
- ELU :- Deep learning models

4) Explain the concept of unsupervised learning in Restricted Boltzmann machines

⇒ Unsupervised Learning :-

It is a method in which the learns patterns from unlabeled data

Restricted Boltzmann Machines

- contains visible and hidden layers
- performs feature extraction and pattern recognition

Autoencoder :-

- contains an encoder and a decoder
- performs data compression and reconstruction

Feature	RBM	Autoencoder
Objective	Feature Extraction	Data reconstruction
Training method	Constructive divergence	Backpropagation
Application	Recommendation System	Image denoising

- 5) Explain the concept of Autoencoder and its uses in DL
- A) Autoencoder is a unsupervised neural network that learn important features from input and reconstructs that original data.

Structure :- flow

Input $\rightarrow$ Encoder $\rightarrow$ Bottleneck layer $\rightarrow$ Decoder $\rightarrow$ output

Uses :-

- Feature Extraction
- Data compression
- Image denoising
- Anomaly detection
- Dimensionality reduction

Structure

- Encoder :- converts the input into compressed form
- Bottleneck layer :- stores the important feature

o Decoder :- Reconstructs the original data

Application :-

o Face recognition

o Medical image analysis

o Recommendation system

o Natural language processing (NLP)